Final Term Project Documentation

<u>Project Name</u>

University Admission Management System

Name	ID	Contribution
Sheikh Rubaeid Sanjid	21-45844-3	(33.34%) Activity Diagram, User Interface, Query Writing
Abdullah Al Faisal	21-45872-3	(33.33%) Project Proposal, Class Diagram, Schema Diagram
Wasifur Rahaman Chowdhury Alvi	21-45856-3	(33.33%) Project Updates, Use case diagram, Relational Algebra

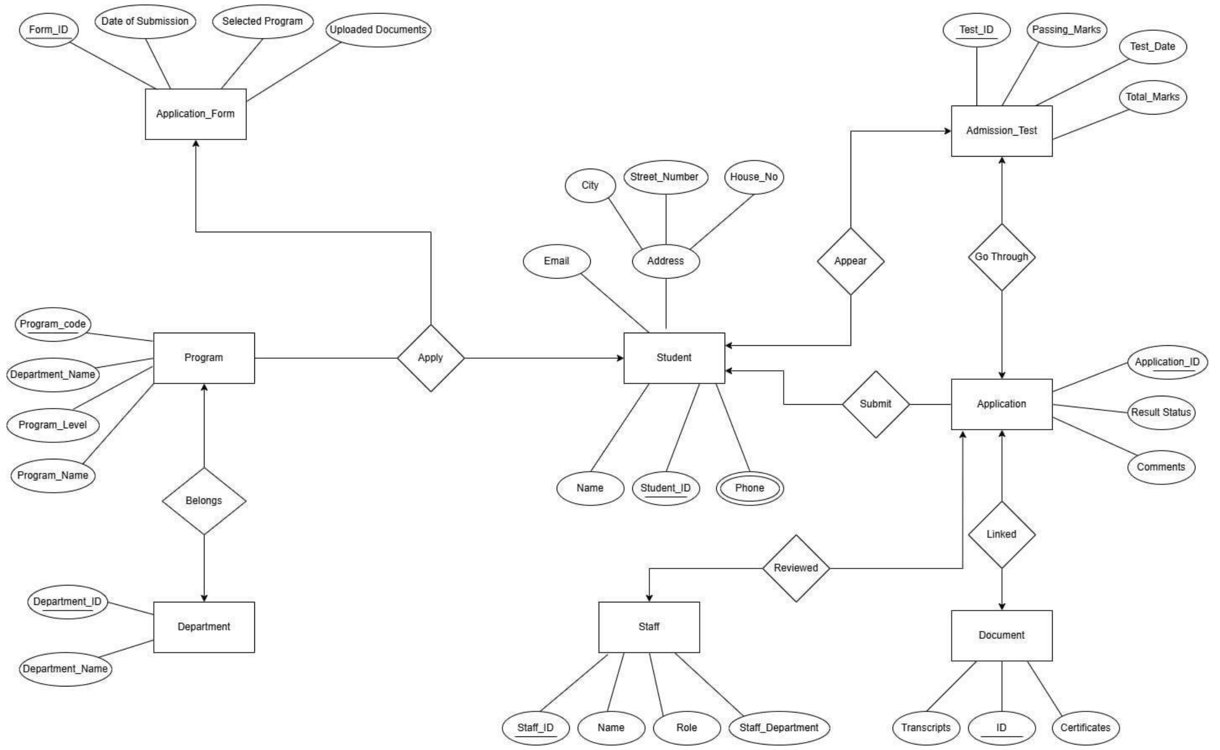
Course Name: Advance Database Management System

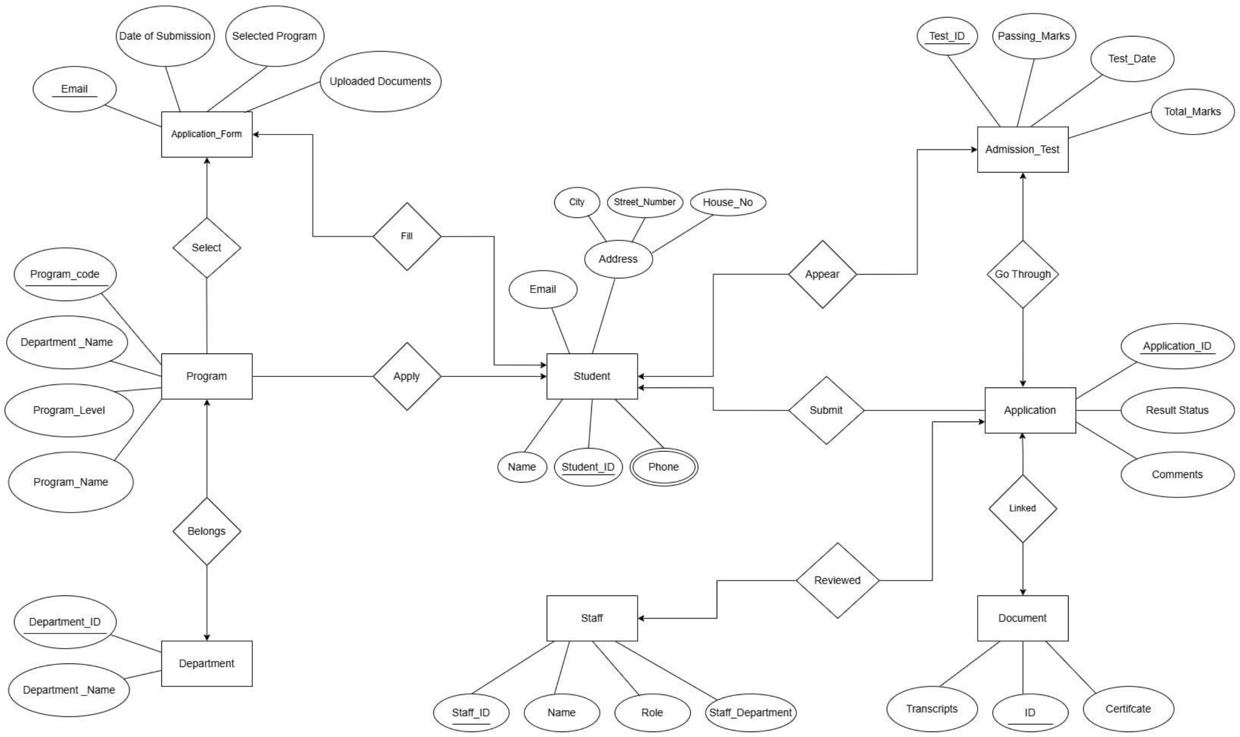
Section: A

Date of Submission: **June 26, 2025**

Project Updates:

ER Diagram:





Updated Normalization:

Apply

UNF

Apply (<u>Student_ID,</u> Name, Phone, Email, City, Street_No , House_No, program_code, department_name, program_level, program_name)

1NF

Phone is a multi valued attribute.

1. <u>Student_ID,</u> Name, Phone, Email, City, Street_No , House_No, <u>program_code</u>, department_name, program_level, program_name

2NF

1. <u>Student_ID,</u> Name, Phone, Email, City, Street_No , House_No,,

2. <u>program_code</u>, department_name, program_level, program_name

3. <u>SFP_Id</u>, Student_ID, program_code

3NF

1. <u>Student_ID,</u> Name, Phone, Email, City, Street_No , House_No,

2. <u>program_code</u>, program_level, **DepPro_Id (FK)**

3. <u>SFP_Id</u>, Student_ID, program_code

4. <u>DepPro_Id</u>, department_name, program_name

Table Creation

1. <u>Student_ID,</u> Name, Phone, Email, City, Street_No , House_No,

2. <u>program_code</u>, program_level, **DepPro_Id (FK)**

3. <u>SFP_Id</u>, Student_ID, program_code

4. <u>DepPro_Id</u>, department_name, program_name

Select

UNF

Select (<u>program_code</u>, department_name, program_level, program_name, <u>email</u>, date_of_submission, selected_program, uploaded_documents)

1NF

There is no multi valued attribute. Relation already in 1NF.

<u>program_code</u>, department_name, program_level, program_name, <u>email</u>, date_of_submission, selected_program, uploaded_documents

2NF

1. <u>program_code</u>, department_name, program_level, program_name

2. <u>email</u>, date_of_submission, selected_program, uploaded_documents

3. <u>program_email_id</u>**,** program_code, email

3NF

1. program_code, program_level, **DepPro_Id (FK),**

2. email, date_of_submission, selected_program, uploaded_documents

3. program_email_id**,** program_code, email

4. DepPro_Id, department_name**,** program_name

Table Creation

1. program_code, program_level, **DepPro_Id (FK),**

2. email, date_of_submission, selected_program, uploaded_documents

3. program_email_id**,** program_code, email

4. DepPro_Id, department_name**,** program_name

FILL

UNF

Fill (Student_ID, Name, Phone, Email, City, Street_No , House_No, email, date_of_submission, selected_program, uploaded_documents)

1NF

Phone is a multi-valued attribute.

Student_ID, Name, Phone, Email, City, Street_No , House_No, email, date_of_submission, selected_program, uploaded_documents

2NF

1. Student_ID, Name, Phone, Email, City, Street_No , House_No

2. email, date_of_submission, selected_program, uploaded_documents

3. student_email_id**,** student_id, email

3NF

There is no transitive dependency. Relation already in 3NF.

1. Student_ID, Name, Phone, Email, City, Street_No , House_No

2. email, date_of_submission, selected_program, uploaded_documents

3. student_email_id**,** student_id, email

Table Creation

1. Student_ID, Name, Phone, Email, City, Street_No , House_No

2. email, date_of_submission, selected_program, uploaded_documents

3. student_email_id**,** student_id, email

Belongs

UNF

Belongs (program_code, department_name, program_level, program_name, department_id, department_name)

1NF

There is no multi valued attribute. Relation already in 1NF.

program_code, department_name, program_level, program_name, department_id, department_name

2NF

1. program_code, department_name, program_level, program_name
2. department_id, department_name
3. DP_Id, program_code, department_id

3NF

1. program_code, program_level, **DepPro_Id (FK)**
2. department_id, department_name
3. DP_Id, program_code, department_id
4. DepPro_Id, department_name, program_name

Table Creation

1. program_code, program_level, **DepPro_Id (FK)**
2. department_id, department_name
3. DP_Id, program_code, department_id
4. DepPro_Id, department_name, program_name

Appear

UNF

Appear (student_id, name, email, phone, city, street_number, house_no, test_id, passing_marks, test_date, total_marks)

1NF

Phone is a multi valued attribute.

1. student_id, name, email, phone, city, street_number, house_no, test_id, passing_marks, test_date, total_marks

2NF

1. student_id, name, email, phone, city, street_number, house_no

2. test_id, passing_marks, test_date, total_marks

3. ST_Id, student_id, test_id

3NF

1. student_id, name, email, phone, city, street_number, house_no

2. test_id, test_date, **PT_Id (FK)**

3. ST_Id**,** student_id, test_id

4. PT_Id, passing_marks, total_marks

Table Creation

1. student_id, name, email, phone, city, street_number, house_no

2. test_id, test_date, **PT_Id (FK)**

3. ST_Id**,** student_id, test_id

4. PT_Id, passing_marks, total_marks

Go Through

UNF

Go Through (application_id, result_status, comments, test_id, passing_marks, test_date, total_marks)

1NF

There is no multi valued attribute. Relation already in 1NF.

1. application_id, result_status, comments, test_id, passing_marks, test_date, total_marks

2NF

1. application_id, result_status, comments,

2. test_id, passing_marks, test_date, total_marks

3. AT_Id**,** application_id, test_id,

3NF

1. application_id, result_status, comments,

2. test_id, test_date, total_marks

3. AT_Id**,** application_id, test_id,

4. PT_Id, passing_marks, total_marks

Table Creation

1. application_id, result_status, comments,

2. test_id, test_date, total_marks

3. AT_Id**,** application_id, test_id,

4. PT_Id, passing_marks, total_marks

Submit

UNF

Submit (application_id, result_status, comments, student_id, name, email, phone, address, city, street_number, house_no)

1NF

Phone is a multi valued attribute.

1. application_id, result_status, comments, student_id, name, email, phone, address, city, street_number, house_no

2NF

1. application_id, result_status, comments

2. student_id, name, email, phone, address, city, street_number, house_no

3. AS_Id**,** application_id, student_id,

3NF

There is no transitive dependency. Relation already in 3NF

1. application_id, result_status, comments

2. student_id, name, email, phone, address, city, street_number, house_no

3. AS_Id**,** application_id, student_id,

Table Creation

1. application_id, result_status, comments

2. student_id, name, email, phone, address, city, street_number, house_no

3. AS_Id**,** application_id, student_id,

Linked

UNF

Linked (application_id, date_of_submission, selected_program, uploaded_documents, document_id, transcripts, certificates**)**

1NF

There is no multi valued attribute. Relation already in 1NF.

1. application_id, date_of_submission, selected_program, uploaded_documents, document_id, transcripts, certificates

2NF

1. application_id, date_of_submission, selected_program, uploaded_documents,

2. document_id, transcripts, certificates

3. AD_Id**,** application_id, document_id,

3NF

There is no transitive dependency. Relation already in 3NF

1. application_id, date_of_submission, selected_program, uploaded_documents,

2. document_id, transcripts, certificates

3. AD_Id**,** application_id, document_id,

Table Creation

1. application_id, date_of_submission, selected_program, uploaded_documents,

2. document_id, transcripts, certificates

3. AD_Id**,** application_id, document_id,

Reviewed

UNF

Reviewed (application_id, result_status, comments, staff_id, name, role, staff_department)

1NF

There is no multi valued attribute. Relation already in 1NF.

1. application_id, result_status, comments, staff_id, name, role, staff_department

2NF

1. application_id, result_status, comments

2. staff_id, name, role, staff_department

3. AS_Id**,** application_id, staff_id

3NF

There is no transitive dependency. Relation already in 3NF

1. application_id, result_status, comments

2. staff_id, name, role, staff_department

3. AS_Id**,** application_id, staff_id

Table Creation

1. application_id, result_status, comments

2. staff_id, name, role, staff_department

3. AS_Id**,** application_id, staff_id

Temporary Tables

1. Student_ID, Name, Phone, Email, City, Street_No , House_No,
2. program_code, program_level, **DepPro_Id (FK)**
3. SFP_Id, Student_ID, program_code
4. DepPro_Id, department_name, program_name
5. program_code, program_level, **DepPro_Id (FK),**
6. email, date_of_submission, selected_program, uploaded_documents
7. program_email_id, program_code, email

8. DepPro_Id, department_name, program_name
9. Student_ID, Name, Phone, Email, City, Street_No , House_No
10. email, date_of_submission, selected_program, uploaded_documents
11. student_email_id, student_id, email
12. program_code, program_level, **DepPro_Id (FK)**
13. department_id, department_name
14. DP_Id, program_code, department_id
15. DepPro_Id, department_name, program_name
16. student_id, name, email, phone, city, street_number, house_no
17. test_id, test_date, **PT_Id (FK)**
18. ST_Id, student_id, test_id
19. PT_Id, passing_marks, total_marks
20. application_id, result_status, comments,
21. test_id, test_date, total_marks
22. AT_Id, application_id, test_id,
23. PT_Id, passing_marks, total_marks
24. application_id, result_status, comments
25. student_id, name, email, phone, address, city, street_number, house_no
26. AS_Id, application_id, student_id,
27. application_id, date_of_submission, selected_program, uploaded_documents,
28. document_id, transcripts, certificates
29. AD_Id, application_id, document_id,
30. application_id, result_status, comments
31. staff_id, name, role, staff_department
32. AS_Id, application_id, staff_id

Final Tables

1. Student_ID, Name, Phone, Email, City, Street_No, House_No,
2. program_code, program_level, **DepPro_Id (FK)**
3. SFP_Id, Student_ID, program_code
4. DepPro_Id, department_name, program_name
5. <u>email</u>, date_of_submission, selected_program, uploaded_documents
6. program_email_id, program_code, email
7. student_email_id, student_id, email
8. department_id, department_name
9. DP_Id, program_code, department_id
10. test_id, test_date, **PT_Id (FK)**

11. ST_Id, student_id, test_id
12. PT_Id, passing_marks, total_marks
13. application_id, result_status, comments,
14. test_id, test_date, total_marks
15. AT_Id, application_id, test_id,
16. student_id, name, email, phone, city, street_number, house_no
17. AS_Id, application_id, student_id,
18. document_id, transcripts, certificates
19. AD_Id, application_id, document_id,
20. staff_id, name, role, staff_department
21. AS_Id, application_id, staff_id

Updated Basic PL/SQL by removing random data:

1. Operators

Question 1:

"Compare if the student's total_marks are greater than the passing marks, and output if they passed or failed."

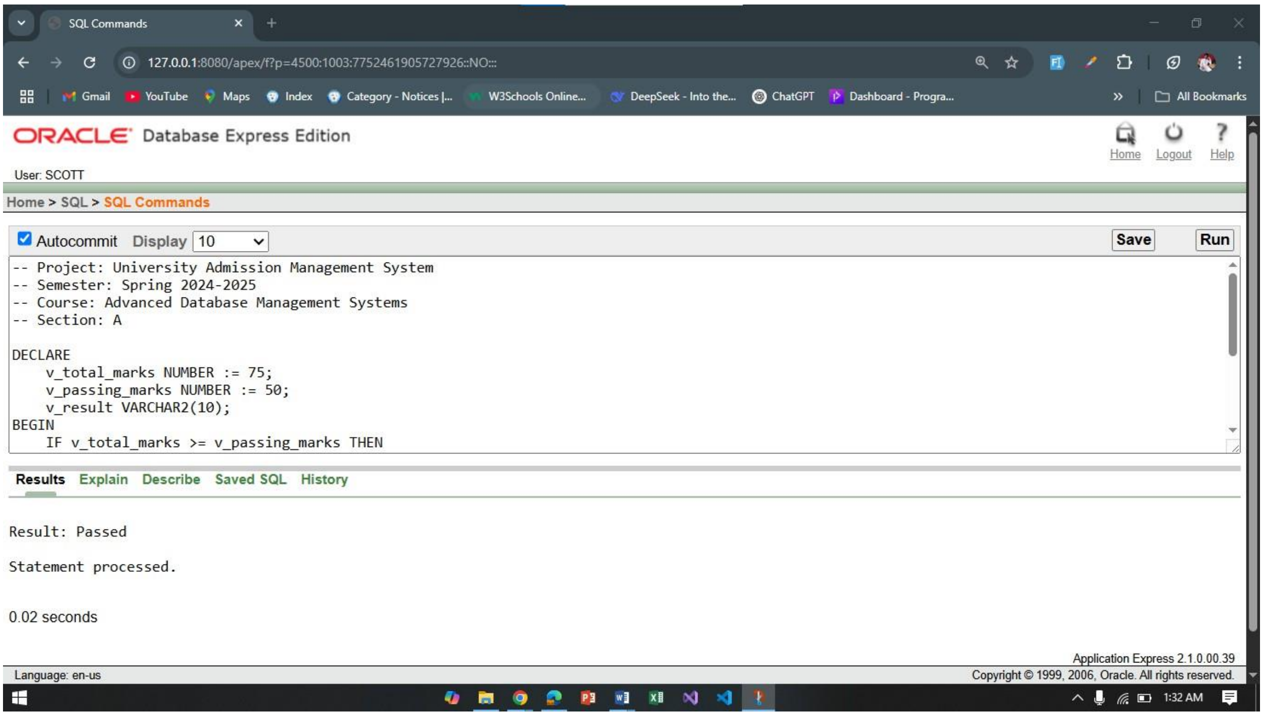
```
-- Project: University Admission Management System
-- Semester: Spring 2024-2025
-- Course: Advanced Database Management Systems
-- Section: A


DECLARE
   v_student_id NUMBER := 1;
   v_test_id NUMBER;
   v_total_marks NUMBER;
   v_passing_marks NUMBER;
   v_result VARCHAR2(10);
BEGIN
   SELECT test_id
   INTO v_test_id
   FROM student_test
   WHERE student_id = v_student_id;

   SELECT t.total_marks, p.passing_marks
   INTO v_total_marks, v_passing_marks
   FROM test t
```

```
    JOIN passing_total p ON t.pt_id = p.pt_id
    WHERE t.test_id = v_test_id;

    IF v_total_marks >= v_passing_marks THEN
        v_result := 'Passed';
    ELSE
        v_result := 'Failed';
    END IF;
    DBMS_OUTPUT.PUT_LINE('Result: ' || v_result);
END;
```



Question 2:

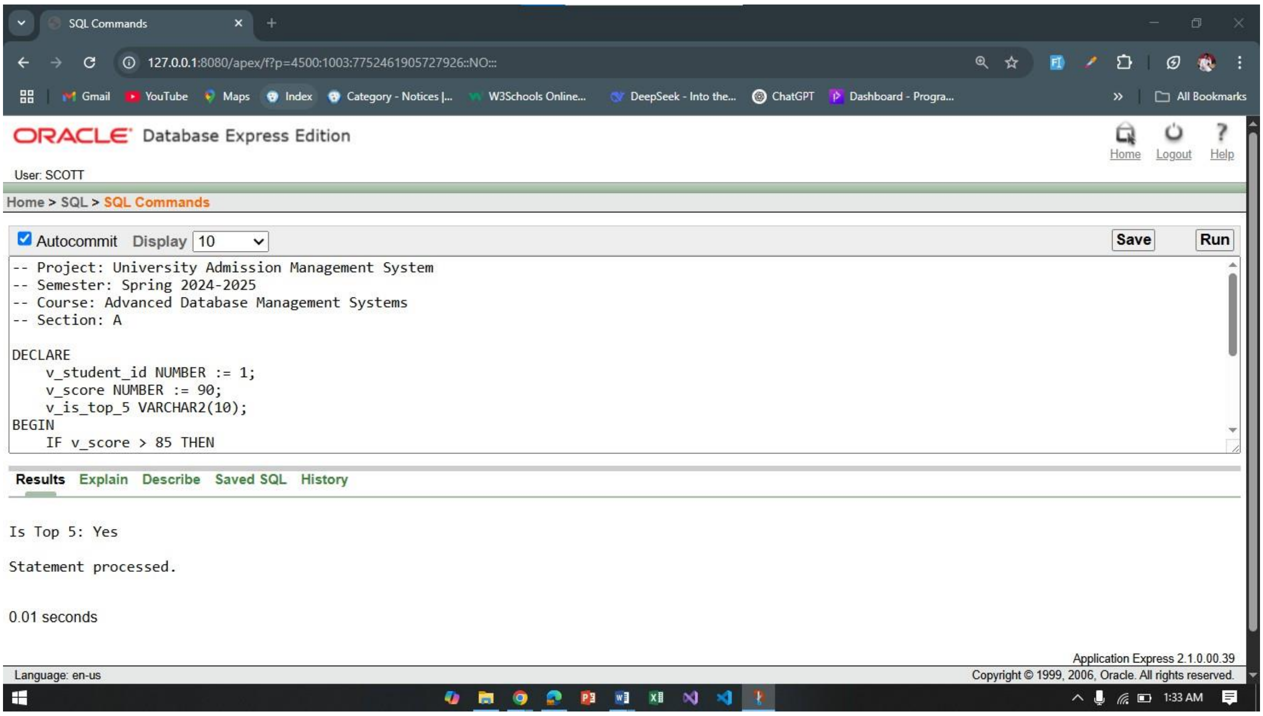
"Check if a student is in the top 5 based on their score."

```
-- Project: University Admission Management System
-- Semester: Spring 2024-2025
-- Course: Advanced Database Management Systems
-- Section: A

DECLARE
    v_student_id NUMBER := 1;
    v_score NUMBER;
    v_is_top_5 VARCHAR2(10);
BEGIN
```

```
    SELECT ST.total_marks
    INTO v_score
    FROM Student_Test ST
    WHERE ST.Student_ID = v_student_id;

    IF v_score >= (
        SELECT MIN(ST2.total_marks)
        FROM (
            SELECT total_marks
            FROM Student_Test
            ORDER BY total_marks DESC
            FETCH FIRST 5 ROWS ONLY
        ) ST2
    ) THEN
        v_is_top_5 := 'Yes';
    ELSE
        v_is_top_5 := 'No';
    END IF;
    DBMS_OUTPUT.PUT_LINE('Is Top 5: ' || v_is_top_5);
END;
```



2. Conditional Statements

Question 2:

"Determine if a student has scored more than 60 marks."

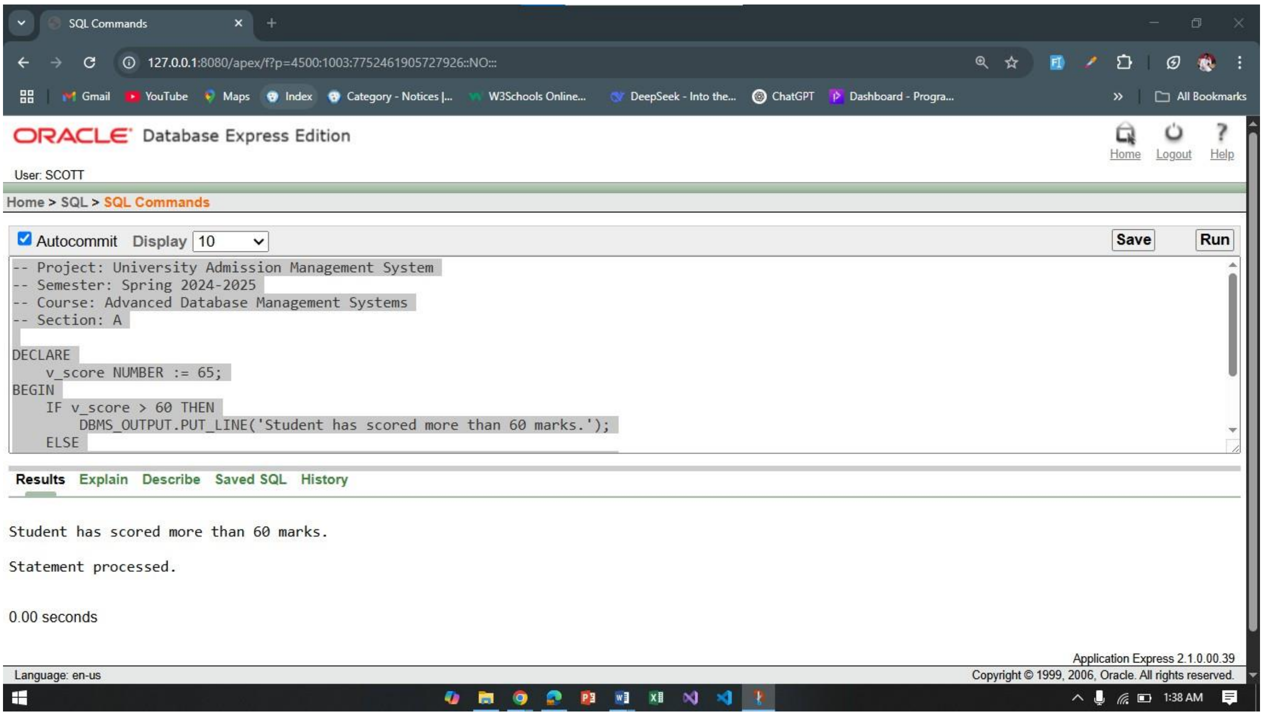
-- Project: University Admission Management System

-- Semester: Spring 2024-2025

-- Course: Advanced Database Management Systems

-- Section: A

```
DECLARE
   v_student_id NUMBER := 1;
   v_score NUMBER;
BEGIN
   SELECT ST.total_marks
   INTO v_score
   FROM Student_Test ST
   WHERE ST.Student_ID = v_student_id;
   IF v_score > 60 THEN
      DBMS_OUTPUT.PUT_LINE('Student has scored more than 60 marks.');
   ELSE
      DBMS_OUTPUT.PUT_LINE('Student has scored less than 60 marks.');
   END IF;
END;
```



Project Proposal:

Overview

The University Admission Management System is designed to simplify and automate the university admission process. This system aims to handle the application process for students, manage student data, and assist university staff in reviewing and processing applications. The system will use a centralized database to store key information such as student details, application forms, program offerings, and admission results. The primary goal of the system is to reduce manual work, minimize errors, and accelerate the admission process.

Project Objectives

The main objectives of the University Admission Management System are:

- **Simplifying the Admission Process:** Providing an online platform where students can easily apply for different programs, fill out personal and academic information, and submit their application forms.
- **Streamlining Data Management:** Storing all student and program data in a safe, organized, and easily accessible database, ensuring data integrity and security.
- **Enhancing Reporting and Monitoring:** Generating reports for administrators to review student applications, track their status, and make informed decisions.
- **Improving Communication and Transparency:** Allowing students to check the status of their applications in real-time, ensuring better communication and transparency throughout the process.
- **Admin Panel for Efficient Management:** Providing an admin panel for university staff to review, approve, or reject applications, making the entire application process more efficient.

Key Features of the System

- **Student Application Form:** Online form for students to submit their personal and academic information. Option to upload supporting documents.
- **Database Management:** Secure and organized storage of all student, program, and application data.
- **Application Status Tracking:** Students can log in to track the status of their applications.
- **Admin Panel:** Admins can review student applications, check completeness, and approve or reject applications.
- **Reporting and Analytics:** Generate statistical reports on application trends, acceptance rates, and other key metrics for university staff.

Benefits of the System

For Students:

- Easy access to the application form and submission process.
- Transparent application status tracking, reducing uncertainty.

For University Staff:

- Streamlined process for managing and reviewing applications.
- Real-time updates and reports for better decision-making and less administrative work.

For the University:

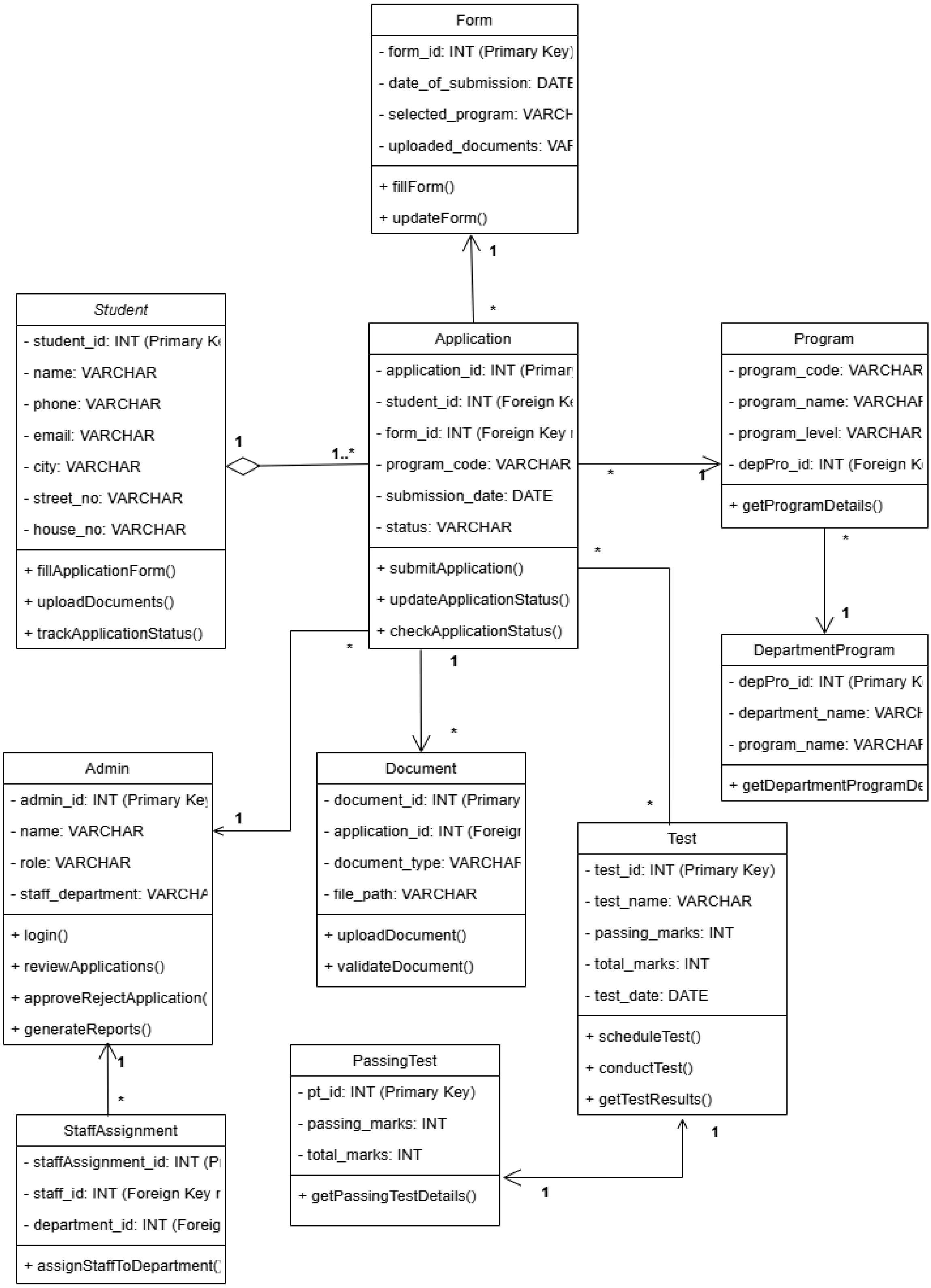
- Reduced errors and human intervention in the admission process.
- Increased efficiency in managing applications and making admission decisions.

Technologies to be Used

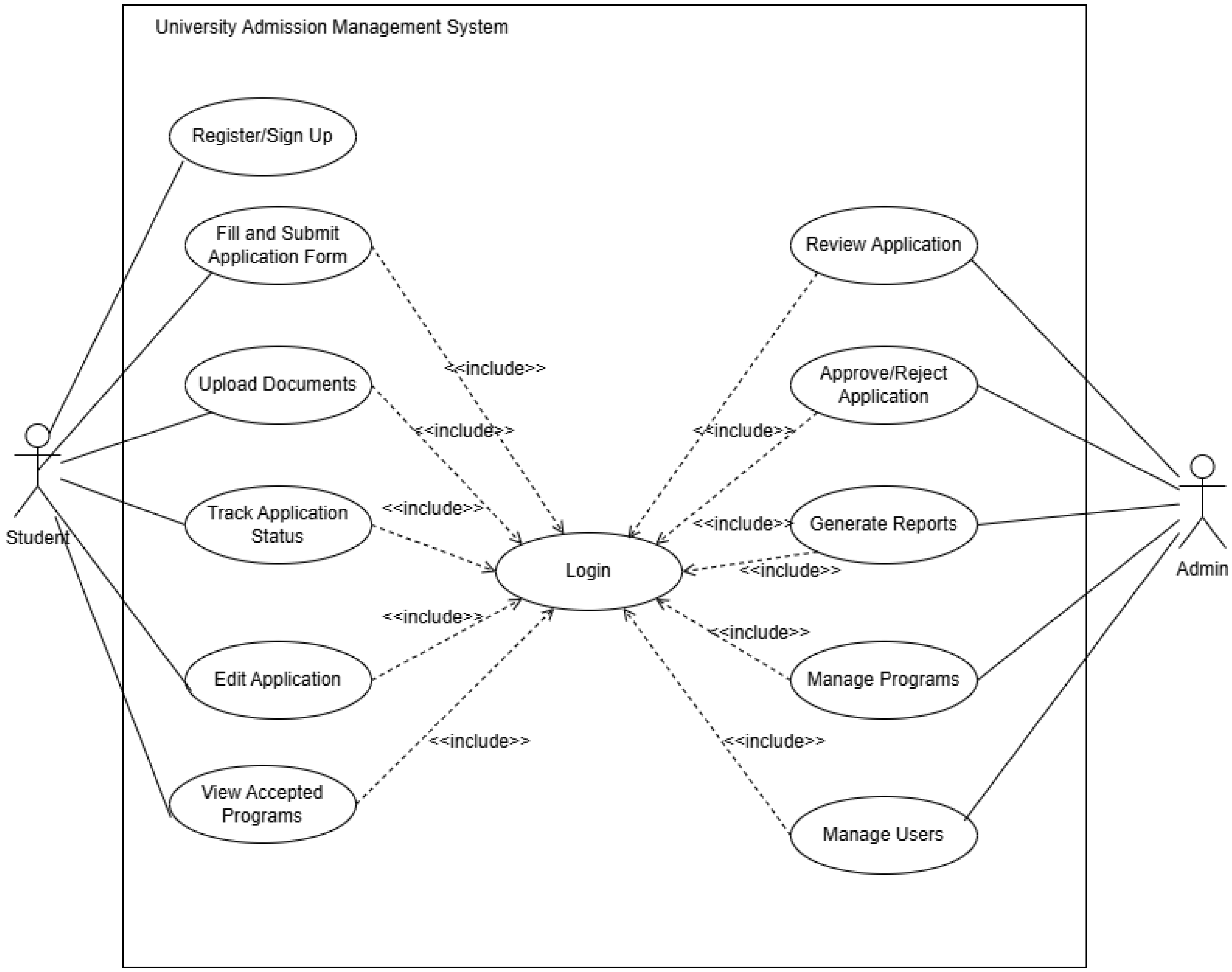
- **Frontend:** HTML, CSS, JavaScript for building the user interface.
- **Backend:** Database management system using Oracle, and server-side using PHP.

Class Diagram, Use Case Diagram, Activity Diagram

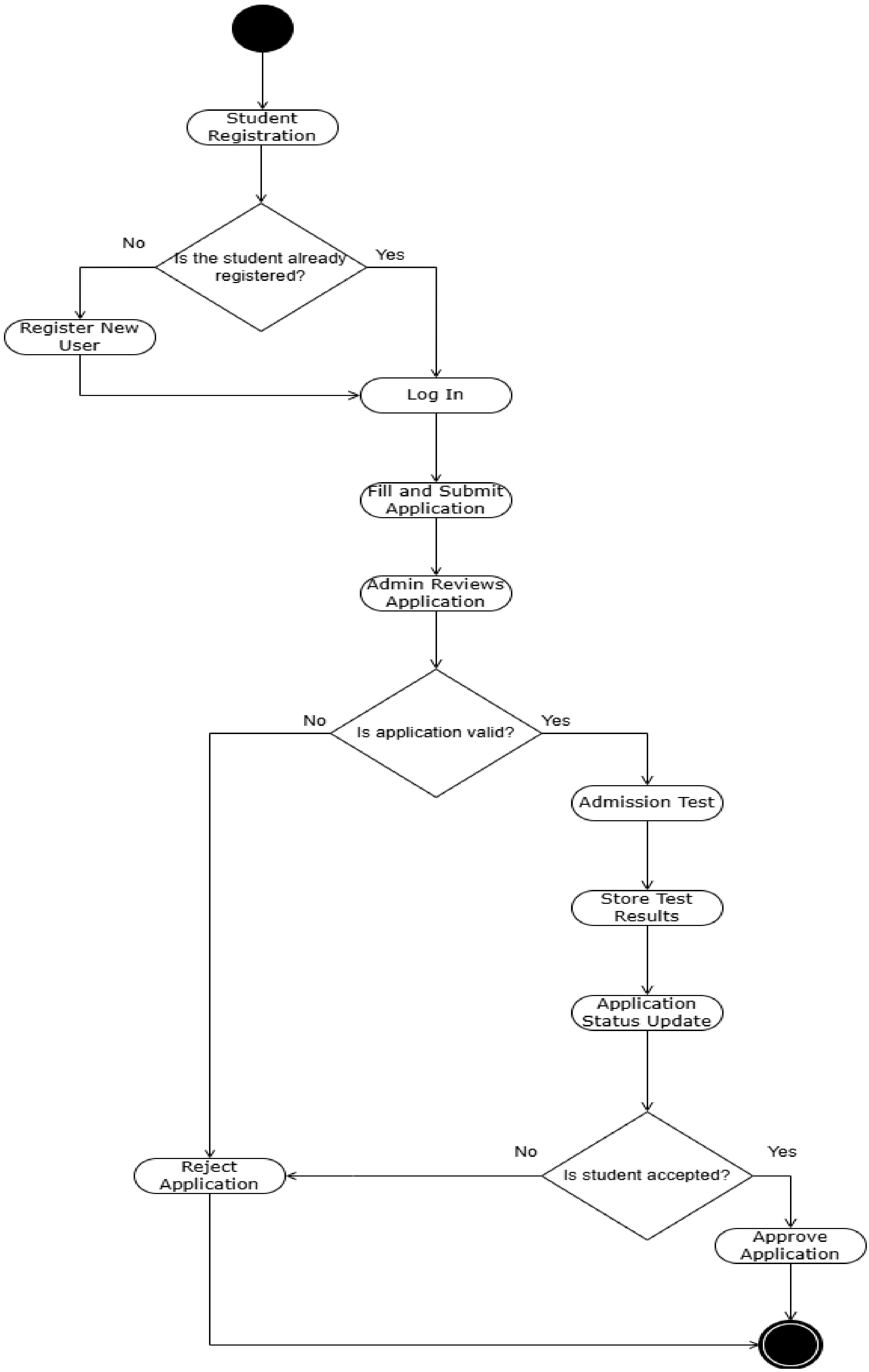
Class Diagram:



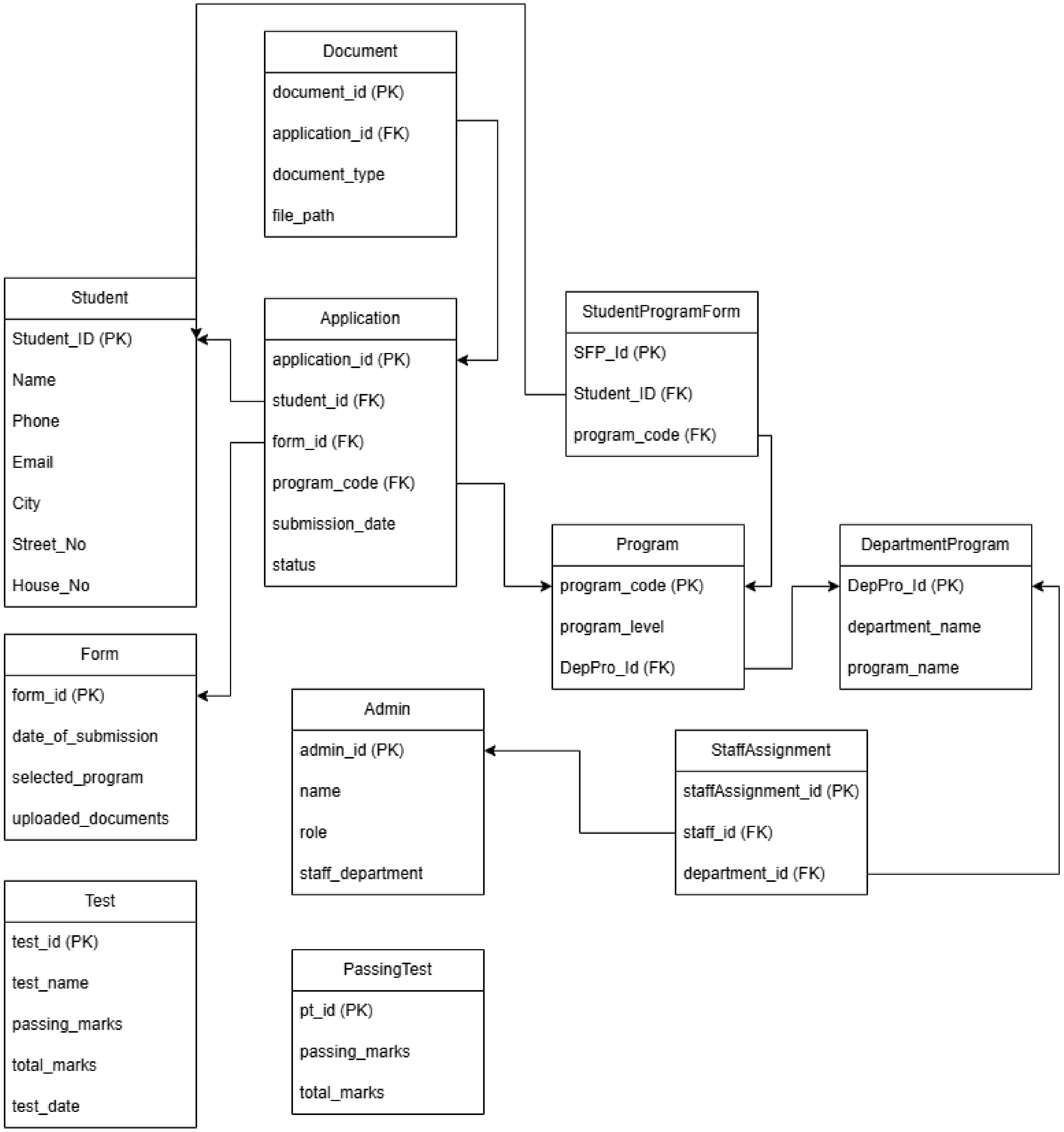
Use Case Diagram:



Activity Diagram:



Schema Diagram:



User Interface:

Mid Term User Interface Design:

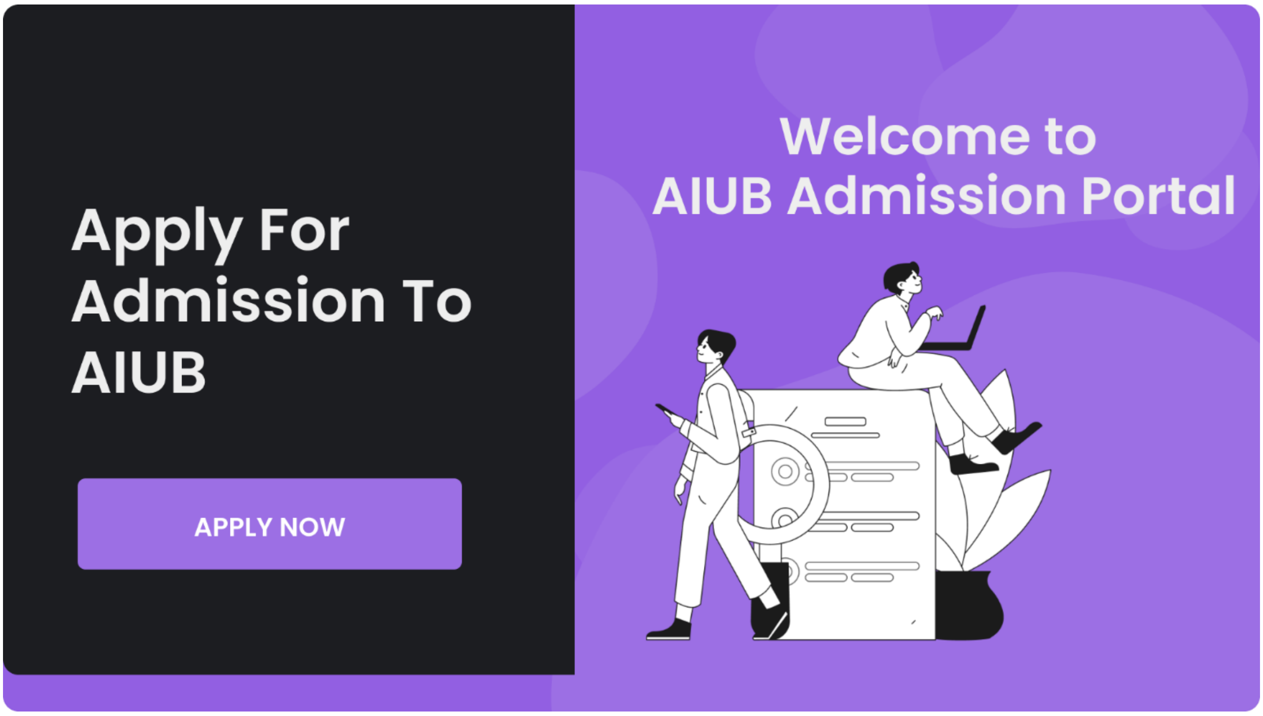


Figure: Landing Page

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>University Admission Management System</title>

    <script src="https://cdn.tailwindcss.com"></script>
</head>
<body class="bg-purple-50">

    <div class="flex items-center justify-center h-screen">
        <div class="w-full max-w-4xl rounded-lg shadow-lg flex p-8">
            <!-- Left Section -->
            <div class="w-1/2 bg-black text-white p-6 rounded-l-lg flex flex-col justify-center items-center">
                <h1 class="text-4xl font-semibold mb-4">Apply For Admission To AIUB</h1>
                <a href="application_form.php" class="bg-purple-500 hover:bg-purple-700 text-white py-2 px-6 rou
            </div>

            <!-- Right Section -->
            <div class="w-1/2 bg-purple-500 text-white p-6 rounded-r-lg flex flex-col justify-center items-cente
                <h2 class="text-4xl font-semibold mb-4">Welcome to AIUB Admission Portal</h2>
                <img src="images/banner.png" alt="AIUB Image" class="w-3/4 mt-4">
            </div>
        </div>
    </div>

</body>
</html>
```

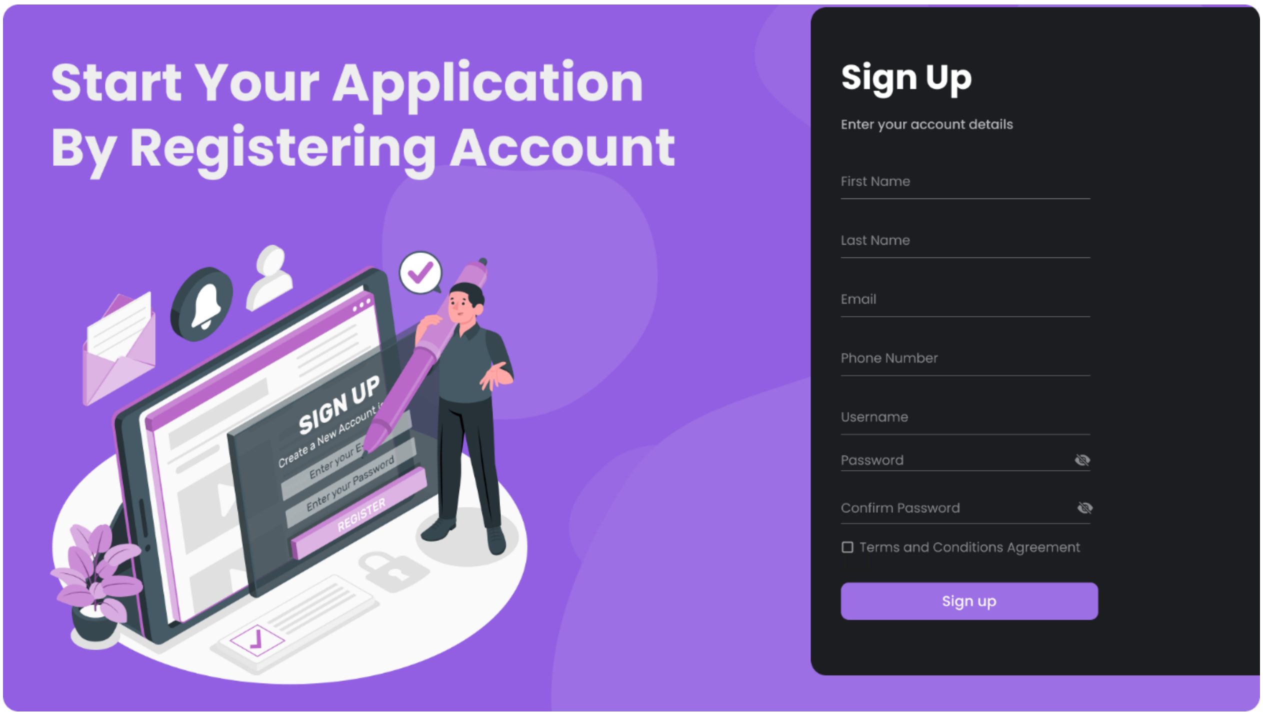


Figure: Signup Page

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Signup - University Admission</title>
    <script src="https://cdn.tailwindcss.com"></script>
    <style>
        body {
            background-color: #F3E8FF;
        }
    </style>
</head>
<body class="bg-purple-50">

    <div class="flex items-center justify-center h-screen">

        <!-- Signup container -->
        <div class="w-full max-w-4xl bg-white rounded-lg shadow-lg p-8 flex">

            <div class="w-1/2 pr-8">
                <img src="images/SignUp.png" alt="Illustration" class="w-full h-auto rounded-lg">
            </div>

            <!-- Right side form -->
            <div class="w-1/2">
                <h1 class="text-4xl font-semibold text-center bg-black text-purple-500 mb-6">Start Your Applicat

                <form id="signupForm" class="space-y-6">

                    <div>
                        <label for="firstName" class="block text-lg text-gray-700">First Name</label>
                        <input type="text" name="firstName" id="firstName" class="w-full border-2 border-gray-30
```

Figure: Login Page

login.php > html > body.bg-purple-50 > div.flex.items-center.justify-center.h-screen > div.w-full.max-w-md.bg-white.rounded-lg.shadow-lg.p-8

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Login - University Admission</title>
    <script src="https://cdn.tailwindcss.com"></script>
</head>
<body class="bg-purple-50">

    <div class="flex items-center justify-center h-screen">
        <div class="w-full max-w-md bg-white rounded-lg shadow-lg p-8">
            <h1 class="text-3xl font-semibold text-center text-purple-500 mb-6">Login</h1>

            <form action="login_process.php" method="POST">
                <div class="mb-4">
                    <label for="username" class="block text-lg">Username</label>
                    <input type="text" name="username" id="username" class="w-full border-2 border-gray-300 roun
                </div>

                <div class="mb-4">
                    <label for="password" class="block text-lg">Password</label>
                    <input type="password" name="password" id="password" class="w-full border-2 border-gray-300
                </div>

                <div class="flex justify-center mt-4">
                    <button type="submit" class="bg-purple-500 hover:bg-purple-700 text-white py-2 px-6 rounded-
                </div>
                <div class="mt-4 text-center">
                    <p>Don't have an account? <a href="signup.php" class="text-purple-600">Sign up</a></p>
                </div>
            </form>
        </div>
```

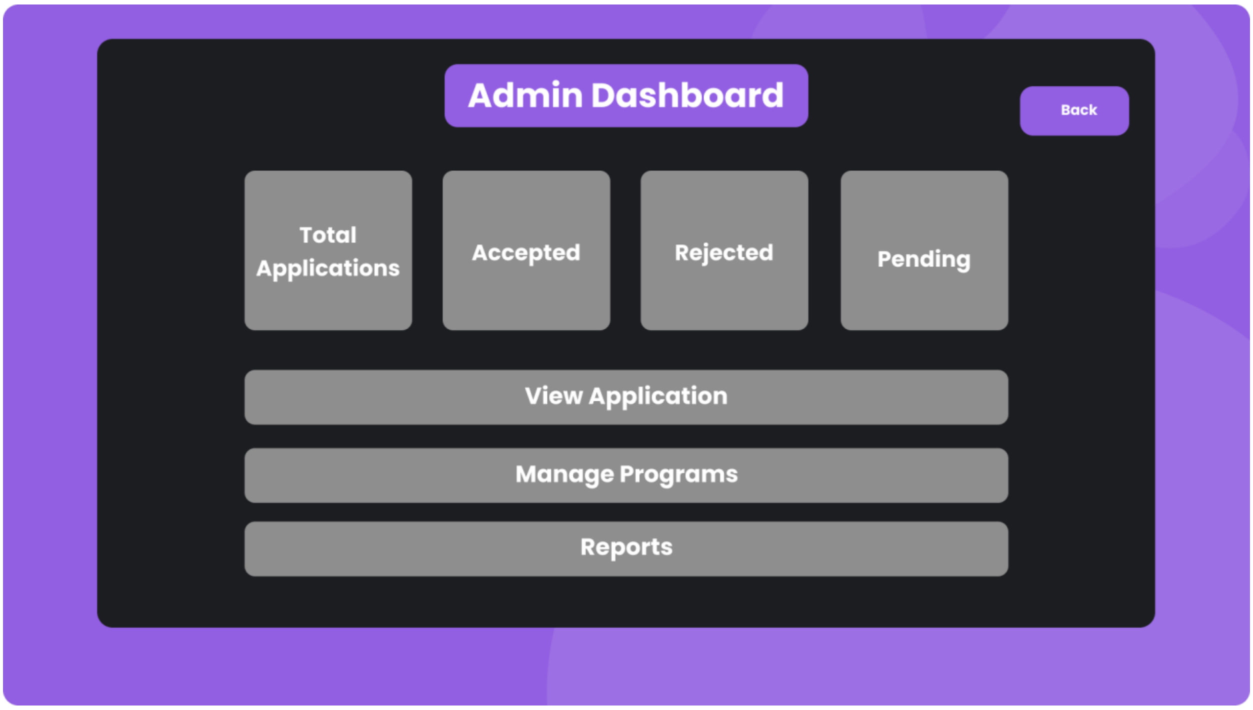


Figure: Admin Panel Page

admin_dashboard.php

admin_dashboard.php > ...

```php
<?php
    session_start();
    if (!isset($_SESSION['user_id']) || $_SESSION['role'] !== 'admin') {
        header('Location: login.php');
        exit();
    }
?>

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Admin Dashboard - University Admission</title>
    <script src="https://cdn.tailwindcss.com"></script>
</head>
<body class="bg-purple-50">

    <div class="flex items-center justify-center h-screen">

        <div class="w-full max-w-4xl bg-white rounded-lg shadow-lg p-8">

            <h1 class="text-3xl font-semibold text-center text-purple-500 mb-6">Admin Dashboard</h1>

            <div class="grid grid-cols-2 lg:grid-cols-4 gap-4 mb-6">
                <div class="bg-purple-500 text-white p-4 rounded-lg text-center">
                    <h2 class="font-semibold">Total Applications</h2>
                    <p class="text-3xl">50</p>
                </div>

                <div class="bg-green-500 text-white p-4 rounded-lg text-center">
                    <h2 class="font-semibold">Accepted</h2>
                    <p class="text-3xl">30</p>
```

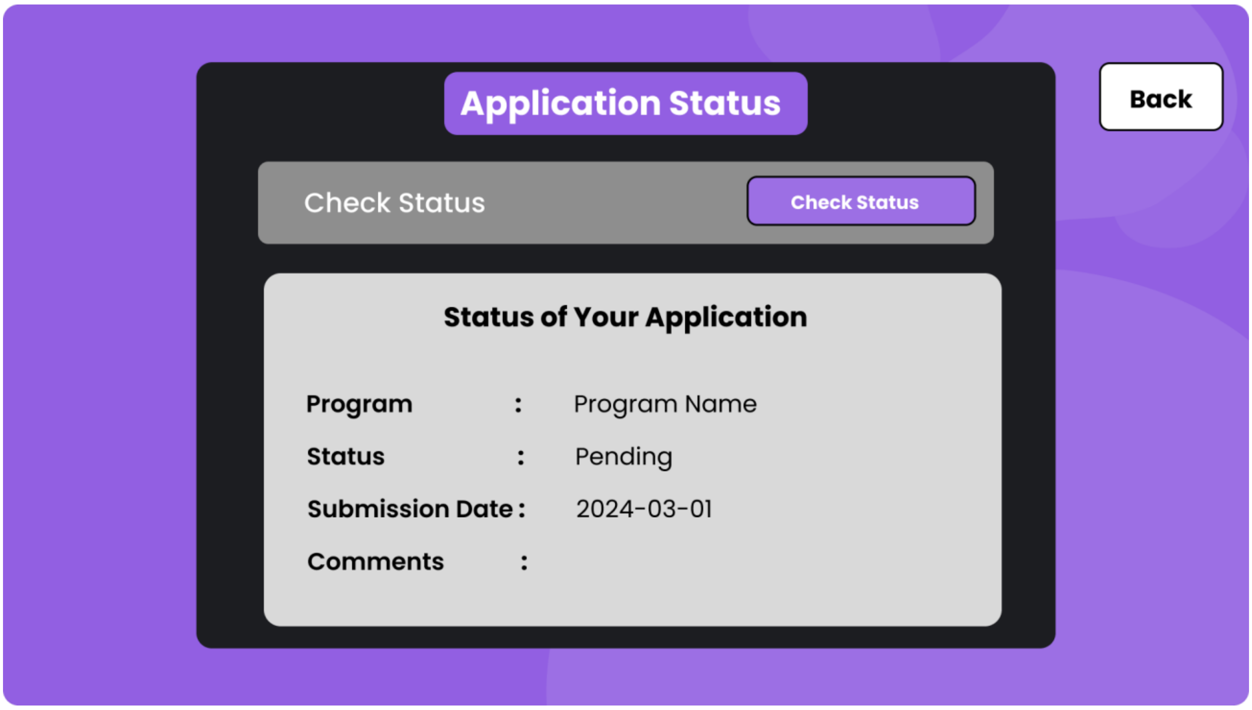


Figure: Application Status Page

application_status.php

```
<?php
session_start();
include('db.php');

if (!isset($_SESSION['user_id'])) {
    header('Location: login.php');
    exit();
}

$userId = $_SESSION['user_id'];
$query = "SELECT * FROM applications WHERE user_id = :userId";
$stmt = oci_parse($oracle_connection, $query);
oci_bind_by_name($stmt, ':userId', $userId);
oci_execute($stmt);
$application = oci_fetch_assoc($stmt);
?>

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Application Status - University Admission</title>
    <script src="https://cdn.tailwindcss.com"></script>
</head>
<body class="bg-purple-50">

    <div class="flex justify-center items-center h-screen">
        <div class="w-full max-w-lg bg-white rounded-lg shadow-lg p-8">
            <h1 class="text-3xl font-semibold text-center text-purple-500 mb-6">Application Status</h1>

            <div class="mb-4">
                <label class="block text-lg">Program:</label>
```

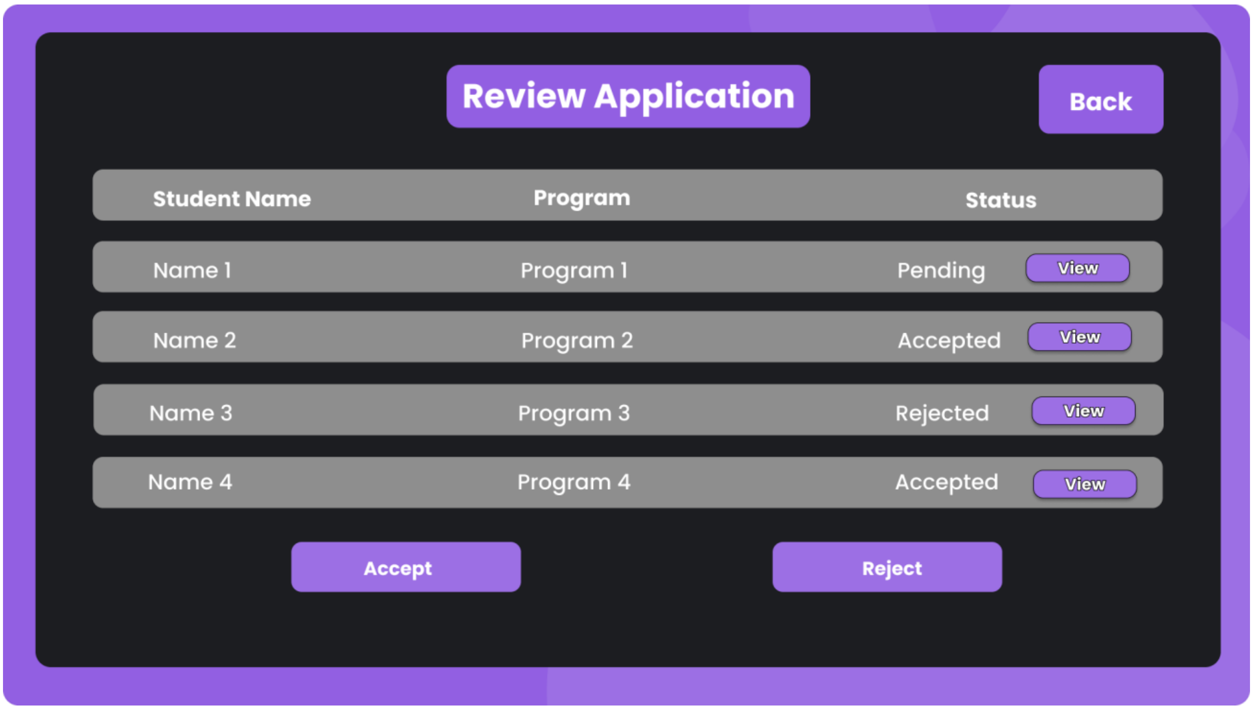


Figure: Review Application Page

view_applications.php ×

view_applications.php > ...

```php
<?php
session_start();
if (!isset($_SESSION['user_id']) || $_SESSION['role'] !== 'admin') {
    header('Location: login.php');
    exit();
}

include('db.php');
$query = "SELECT * FROM applications";
$stmt = oci_parse($oracle_connection, $query);
oci_execute($stmt);
?>

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>View Applications - Admin</title>
    <script src="https://cdn.tailwindcss.com"></script>
</head>
<body class="bg-purple-50">

    <div class="flex justify-center items-center h-screen">
        <div class="w-full max-w-4xl bg-white rounded-lg shadow-lg p-8">
            <h1 class="text-3xl font-semibold text-center text-purple-500 mb-6">Review Applications</h1>

            <table class="min-w-full bg-white">
                <thead>
                    <tr>
                        <th class="py-2 px-4 text-left">Student Name</th>
                        <th class="py-2 px-4 text-left">Program</th>
                        <th class="py-2 px-4 text-left">Status</th>
```

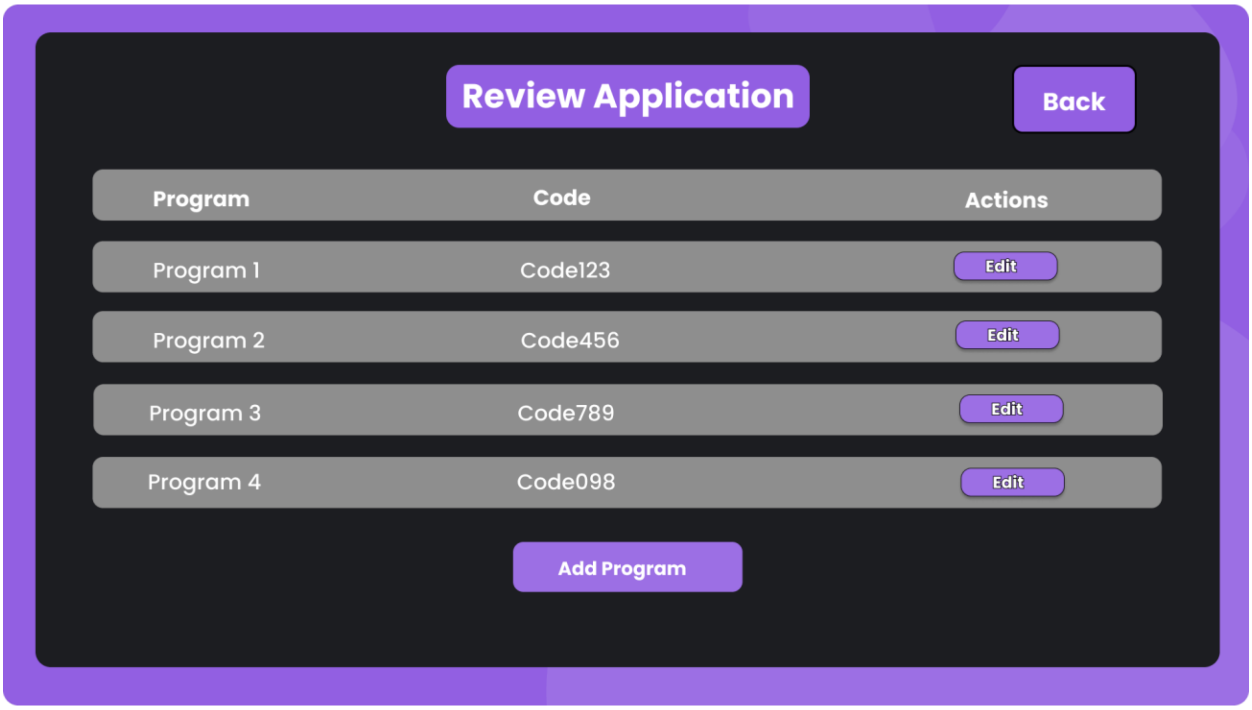


Figure: Review Page

review_application_program.php ×

review_application_program.php > html > body.bg-purple-50

```php
    header('Location: login.php');
    exit();
}

$query = "SELECT * FROM program";
$stmt = oci_parse($oracle_connection, $query);
oci_execute($stmt);
?>

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Review Applications - University Admission</title>
    <script src="https://cdn.tailwindcss.com"></script>
</head>
<body class="bg-purple-50">

    <div class="flex justify-center items-center h-screen">
        <div class="w-full max-w-4xl bg-white rounded-lg shadow-lg p-8">
            <h1 class="text-3xl font-semibold text-center text-purple-500 mb-6">Review Application</h1>

            <table class="min-w-full table-auto mb-6">
                <thead>
                    <tr class="bg-gray-700 text-white">
                        <th class="py-2 px-4 text-left">Program</th>
                        <th class="py-2 px-4 text-left">Code</th>
                        <th class="py-2 px-4 text-center">Actions</th>
                    </tr>
                </thead>
                <tbody>
                    <?php while ($program = oci_fetch_assoc($stmt)): ?>
```

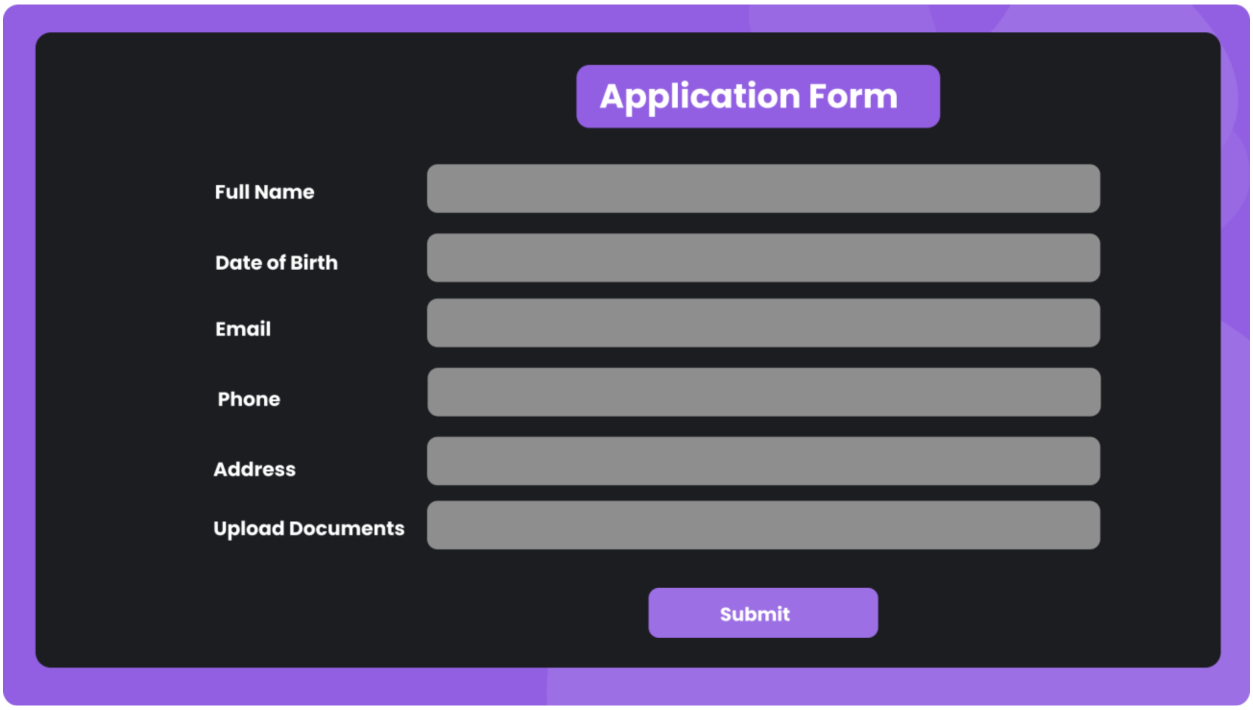


Figure: Application Form Page

```
application_form.php ×
application_form.php > ...
        header('Location: login.php');
        exit();
    }
    ?>

    <!DOCTYPE html>
    <html lang="en">
    <head>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
        <title>Application Form - University Admission</title>
        <script src="https://cdn.tailwindcss.com"></script>
    </head>
    <body class="bg-purple-50">

        <div class="flex justify-center items-center h-screen">
            <div class="w-full max-w-lg bg-white rounded-lg shadow-lg p-8">
                <h1 class="text-3xl font-semibold text-center text-purple-500 mb-6">Application Form</h1>

                <form action="submit_application.php" method="POST" enctype="multipart/form-data">
                    <!-- Full Name -->
                    <div class="mb-4">
                        <label for="fullName" class="block text-lg">Full Name</label>
                        <input type="text" name="fullName" id="fullName" class="w-full border-2 border-gray-300 roun
                    </div>

                    <!-- Date of Birth -->
                    <div class="mb-4">
                        <label for="dob" class="block text-lg">Date of Birth</label>
                        <input type="date" name="dob" id="dob" class="w-full border-2 border-gray-300 rounded-lg py-
                    </div>
```

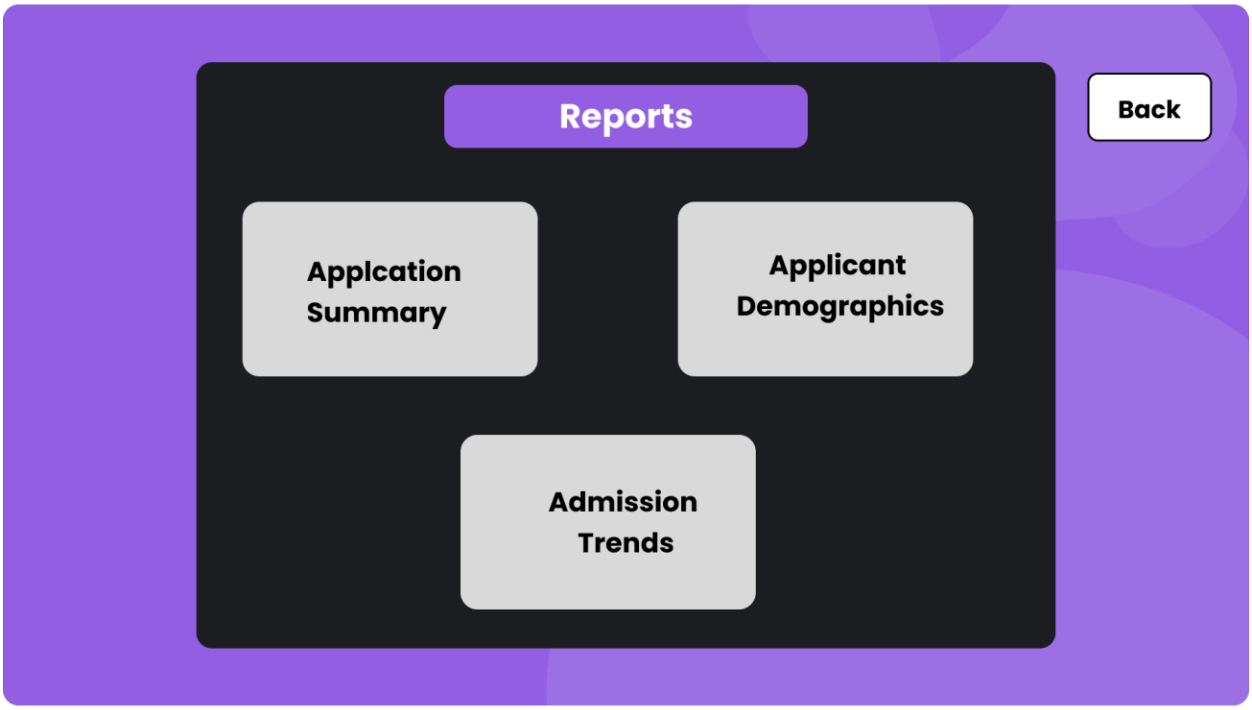


Figure: Reports Page

reports.php

reports.php > ...

```
<?php
session_start();
if (!isset($_SESSION['user_id']) || $_SESSION['role'] !== 'admin') {
    header('Location: login.php');
    exit();
}

include('db.php');
?>

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Reports - Admin</title>
    <script src="https://cdn.tailwindcss.com"></script>
</head>
<body class="bg-purple-50">

    <div class="flex justify-center items-center h-screen">
        <div class="w-full max-w-lg bg-white rounded-lg shadow-lg p-8">
            <h1 class="text-3xl font-semibold text-center text-purple-500 mb-6">Reports</h1>

            <div class="grid grid-cols-1 gap-6">
                <!-- Application Summary Report -->
                <a href="application_summary_report.php" class="bg-purple-500 text-white text-center py-3 rounde

                <!-- Applicant Demographics Report -->
                <a href="applicant_demographics_report.php" class="bg-purple-500 text-white text-center py-3 rou

                <!-- Admission Trends Report -->
                <a href="admission_trends_report.php" class="bg-purple-500 text-white text-center py-3 rounded-l
```

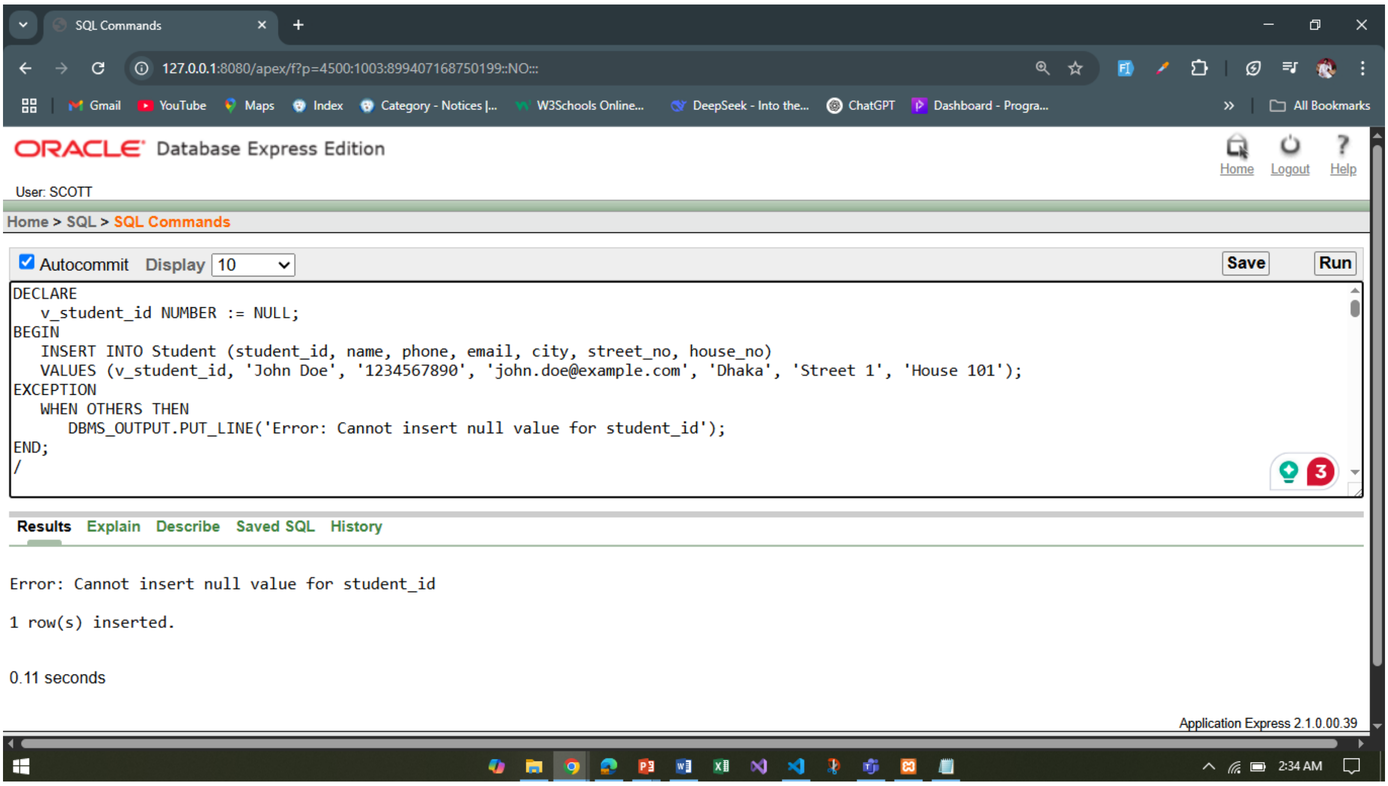
<u>Database Connection</u>:

Query Writing:

Exception Handling

1: Exception Handling - Null Value Insertion

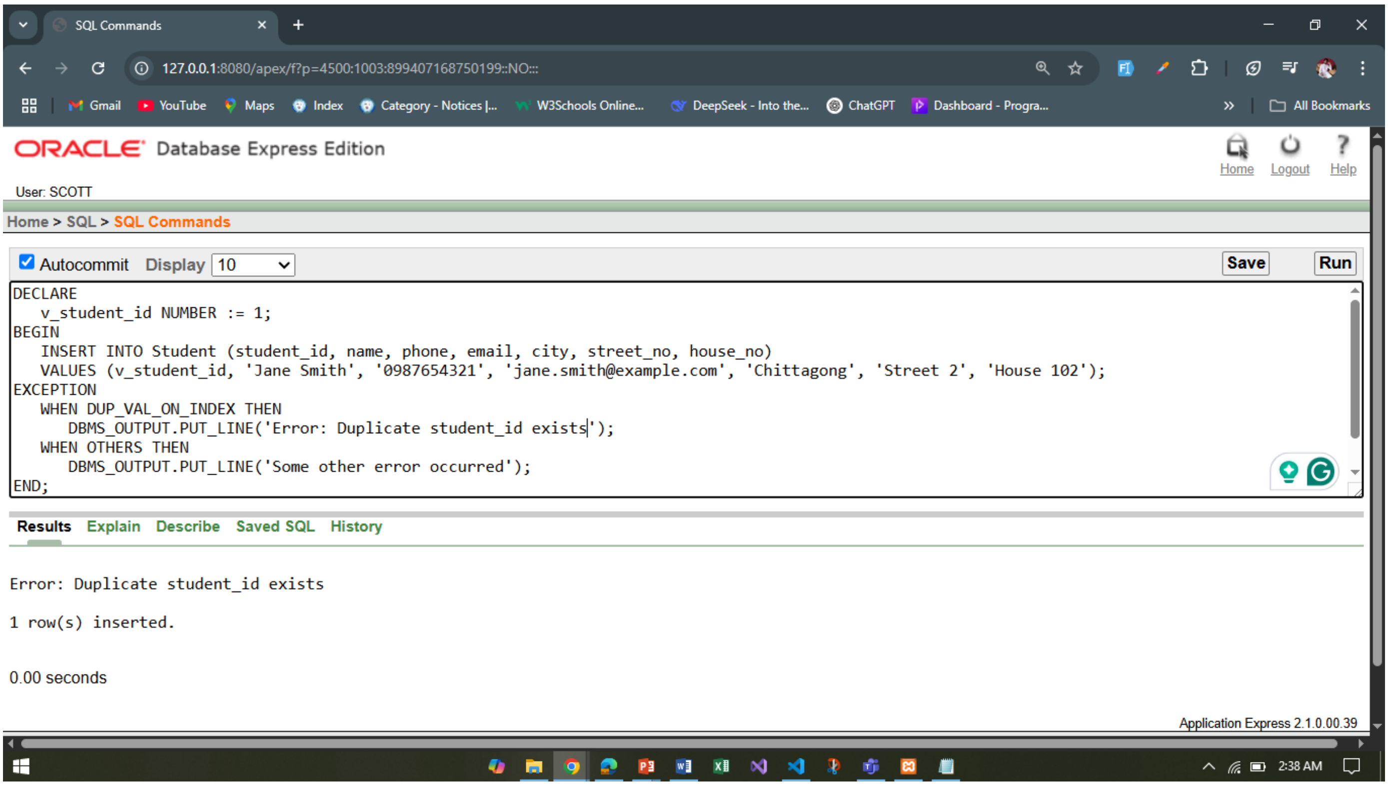
```
DECLARE
  v_student_id NUMBER := NULL;
BEGIN
  INSERT INTO Student (student_id, name, phone, email, city, street_no, house_no)
  VALUES (v_student_id, 'John Doe', '1234567890', 'john.doe@example.com', 'Dhaka', 'Street 1', 'House 101');
EXCEPTION
  WHEN OTHERS THEN
    DBMS_OUTPUT.PUT_LINE('Error: Cannot insert null value for student_id');
END;
/
```



2: Exception Handling - Duplicate Entry

```
DECLARE
  v_student_id NUMBER := 1;
BEGIN
  INSERT INTO Student (student_id, name, phone, email, city, street_no, house_no)
  VALUES (v_student_id, 'Jane Smith', '0987654321', 'jane.smith@example.com', 'Chittagong', 'Street 2', 'House
102');
EXCEPTION
  WHEN DUP_VAL_ON_INDEX THEN
    DBMS_OUTPUT.PUT_LINE('Error: Duplicate student_id exists');
  WHEN OTHERS THEN
    DBMS_OUTPUT.PUT_LINE('Some other error occurred');
```

END;
/



Implicit Locking

1: Implicit Locking - Update Student's Phone Number

```
BEGIN
  UPDATE Student
  SET phone = '9876543210'
  WHERE student_id = 1;
  COMMIT;
END;
/
```

2: Implicit Locking - Delete Student Record

```
BEGIN
  DELETE FROM Student
  WHERE student_id = 2;
  COMMIT;
END;
/
```

Explicit Locking

1: Explicit Locking - Locking a Student Row for Update

```
BEGIN
  SELECT student_id, name
  FROM Student
```

```
   WHERE student_id = 1
   FOR UPDATE;

   UPDATE Student
   SET phone = '1234567890'
   WHERE student_id = 1;
   COMMIT;
END;
/
```

2: Explicit Locking - Locking Multiple Rows

```
BEGIN
   SELECT student_id, name
   FROM Student
   WHERE city = 'Dhaka'
   FOR UPDATE;

   UPDATE Student
   SET phone = '1112223333'
   WHERE city = 'Dhaka';
   COMMIT;
END;
/
```

Relational Algebra

1. Retrieve the names and emails of students who have been accepted:

Answer: π name, email (σ result_status = 'Accepted' (Application ⋈ Student))

2. Find all programs offered by the 'Computer Science' department:

Answer: π program_name (σ department_name = 'Computer Science' (DepartmentProgram))

3. Retrieve the names of students who applied to the 'Data Science' program:

Answer: π name (σ selected_program = 'Data Science' (Application ⋈ Student))

4. Get the list of all students who have submitted documents:

Answer: π student_id (Application ⋈ Document)

5. Find the average total marks of students who appeared in the admission test:

Answer: AVG(total_marks) (StudentTest ⋈ Test)